

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 244

THE SHAKEDOWN DOCTRINE

THE MOON DAY TRIP

not Mars — the ultimate reality test
whole ship, all equipment, down to the landers



8 hours to the Moon at sprint — Mars is 47 days

MOON · HOME · VIA MARS — SEND THE ROCKS, KEEP GOING

THE PLAN IS NEVER FINISHED

Mission doctrine — v1.0 in force

GP-IM-244

Revision v1.0 — 24 August 2026

245 of 290

VETTED BATCH 34 — PASS · TRAJECTORY VALIDATION

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 244

PHASE 244: THE SHAKEDOWN DOCTRINE — STATUS: CURRENT — PASS AS A TEST PHILOSOPHY / TRAJECTORY VALIDATION (VET BATCH 34). We are not colonizing Mars — this ship is the ultimate reality test before that is ever attempted. The shakedown is the Moon day trip: whole ship, all equipment, down to the landers, run as a full-dress rehearsal at the file's own speeds. The arithmetic at sprint 30,000 mph (13.4 km/s, 229's number): Moon 384,400 km in about 8 hours; Mars at closest approach is 47 days one-way — the Moon is the only reasonable real test before the hundred years. Amendment 1 seats the departure route: Moon shakedown, home and fix, depart via Mars for full-dress ops again; send the rocks, keep going — the only sample return a non-stop ship gets; the crater mirror is the only sun at a permanently shadowed crater; the sling at the Moon capped at 378 m/s on mild steel. The rule from 225 at fleet scale: the plan is never finished.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-244, v1.0 — 24 August 2026. The two hundred and forty-fifth manual of the program — 245 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the shakedown law as seated (contents line, the doctrine, the arithmetic, the test card — Amendment 1, the departure route: Moon, home, via Mars) — §2 the physics, graded — §3 the system in the shakedown — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 244: **The Shakedown Doctrine (Not Mars — the Moon Day Trip; Whole Ship, All Equipment, Down to the Landers; the Only Reasonable Real Test Before the Hundred Years).** The arithmetic: sprint 13.4 km/s — Moon ~8 hours one-way (a literal day trip); slow cruise ~28 hours; Mars ~47 days at the same speed — months, a mission not a test. The four-beat loop flown on hardware.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

We are not colonizing mars, this ship is the ultimate reality test before that is ever attempted, the moon maybe, but i need an estimate of time at our speed to use it as a test run before departing for 100 years, whole ship all equipment down to landers etc day trip and back to earth. it is simply the only reasonable real test.

- THE DOCTRINE, SEATED: Mars colonization is NOT the mission; the Moon MAYBE — as the test site. The ship itself is the ultimate reality test, and the only reasonable real test is the full-dress run: whole ship, all equipment, down to the landers, there and back, before the hundred years.
- THE ARITHMETIC AT THE FILE'S OWN SPEEDS: sprint 30,000 mph (13.4 km/s — 229's number) — Moon 384,400 km $\approx$ 8 HOURS one-way: eight out, ops, eight back, a LITERAL day trip. Slow cruise 8,500 mph (3.8 km/s — his sat-speed number) $\approx$ 28 hours one-way: a long weekend with full landing ops. Crew-comfort acceleration (~ 1 g) reaches sprint in ~ 23 minutes — negligible against cruise; the day trip stands. THE MOON IS THE ONLY BODY IN THE SKY THAT MAKES 'DAY TRIP' TRUE.
- WHY NOT MARS, IN THE SAME ARITHMETIC: closest approach 54.6 million km $\approx$ 47 DAYS one-way at sprint (first-cut: straight line, constant speed, before trajectory and acceleration budgets make it longer). A Mars 'test' is months — not a test, a mission, flown before the testing that must come first. The doctrine is arithmetic, not preference.
- THE TEST CARD (what a shakedown proves that nothing else can): whole-ship integration under real mission load; landers cycled FOR REAL — descent, surface ops, ascent, re-dock; 237's beam machines assembled in vacuum; 240's catch covers serviced with gloves on; 235's loop run closed; 238's capability matrix verified at mission tempo; 242's mini-moons seeded as lane markers for the ride home. Every phase since 162 either gets tested or gets a note saying why not.
- THE RULE FROM 225 AT FLEET SCALE: the plan is never finished — the shakedown is the four-beat loop flown on hardware: observe the plan in action, then ask was this the best tomato cut. The Moon run is the observation; the hundred-year ship is what survives it. Seated as spoken: 'it is simply the only reasonable real test.'

PHASE 244 — AMENDMENT 1: THE DEPARTURE ROUTE (MOON, HOME, VIA MARS; SEND THE ROCKS, KEEP GOING; THE CRATER'S ONLY SUN; THE SLING'S HONEST NUMBERS) (VERBATIM)

so moon test and back, we leave and go via mars, do the same thing, get them a hug bucket of rocks and send them back into earth orbit for collection and continue on, if there is a crater with or without ice we could test the mirror and hub so we can let them know for asteroid usage, try out the sling at the moon etc etc.

- THE SEQUENCE SEATED: Moon shakedown → home and fix → depart → VIA MARS → full-dress ops again → the rocks go home → the ship continues. 'Via Mars' is legitimate navigation, not tourism: a flyby on the outbound leg is a modest but real gravity assist (Mars caps the kick at ~5 km/s escape-class) and the only mid-route full-dress rehearsal on the itinerary — deep-space ops tested where coming home is NOT a day trip, while Earth is still reachable if the tomato question goes badly.
- THE HUGE BUCKET OF ROCKS — AFFIRMED: send the rocks, keep going — the only sample return a non-returning ship can fly. THE HONEST LINE: the pod pays the return fare the ship won't — its own propulsion and entry system (or aerocapture) into Earth orbit, Earth-side collection waiting. The ship sheds a capsule, not a crew.
- THE CRATER MIRROR — THE ONLY SUN: at a permanently shadowed crater the mirror is not a tool, it is the only sunlight the ice gets. WITH ice: the harvest test (reflect in, sublimate, capture); WITHOUT ice: the sintering test (225's sun-over-the-drill). 'Let them know for asteroid usage' — the Moon validates the mirror-and-hub asteroid kit BEFORE the asteroids: 244's doctrine doing its job one rock early.
- THE SLING AT THE MOON — AFFIRMED AS A HARDWARE TEST, WITH THE NUMBERS: mild steel caps ~378 m/s; low lunar orbit wants ~1.7 km/s, escape ~2.4 km/s. The lunar sling run validates the MACHINE — spin-up, release timing, the pairing law, accuracy — and serves short-haul surface logistics. Throwing rocks all the way to Earth collection is beyond mild steel: that chapter belongs to the hub and whatever scales past it. The seated clause rides: MOON = MOVER MODE ONLY (229 Amendment 2), no atmosphere cleanup on this side of the test.
- 'etc etc' SEATED AS THE OPEN REGISTER: the departure route is the shakedown's second lap, and the register of what gets tested along it stays open.

INSTRUCTION STATUS (10 August 2026, sitting 147): PARTIAL — the author's voice seated in this body is the surviving fragment of the instruction; the full live order and its thread were not preserved (the 200–260 band was rebuilt by the audit without the chat record). The author counts this phase among the 80 lost instructions. The seated voice is the instruction of record; the audit prose around it is scaffolding.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DOCTRINE, SEATED: Mars colonization is NOT the mission; the Moon MAYBE — as the test site. The ship itself is graded as the ultimate reality test: the whole machine, all equipment, down to the landers, run hot before the hundred years begin. A test you can abort home from is the only honest rehearsal for a test you cannot.

THE ARITHMETIC AT THE FILE'S OWN SPEEDS: sprint 30,000 mph (13.4 km/s — 229's number) puts the Moon 384,400 km away at about 8 hours — a day trip. The same arithmetic prices Mars: closest

approach 54.6 million km is about 47 days one-way at sprint. Why not Mars is not philosophy, it is arithmetic.

THE TEST CARD — WHAT A SHAKEDOWN PROVES THAT NOTHING ELSE CAN: whole-ship integration under real flight; every lander flown, every drill run, every system load-tested in the field it was built for; the crew worked as a crew. Nothing on the ground integrates the machine — only the day trip does.

THE DEPARTURE ROUTE, SEATED (Amendment 1): Moon shakedown, home and fix, depart — VIA MARS — full-dress ops again at the far planet before the long dark. The shakedown is the four-beat loop flown as a route: test, fix, deeper test, go.

THE ROCKS, AFFIRMED: send the rocks, keep going — the only sample return a non-stop ship gets. The huge bucket of rocks goes home by throw; the ship does not turn for them. Science without a course change.

THE CRATER MIRROR AND THE SLING, PRICED: at a permanently shadowed crater the mirror is not a tool, it is the only sun — the Archimedes array's hardest posting. The sling at the Moon is affirmed as a hardware test with the numbers: mild steel caps at about 378 m/s — the Moon's shallow well makes it a service-station exercise, not a siege.

THE RULE FROM 225 AT FLEET SCALE: the plan is never finished — the shakedown is the four-beat loop itself, and 'etc etc' is seated as the open register: the departure route is the shakedown's second lap, and the list stays open.

§3 — THE SYSTEM IN THE SHAKEDOWN

- **The day trip:** Moon at about 8 hours on sprint — whole ship, all equipment, down to the landers.
- **The arithmetic:** Mars at 47 days one-way is why the rehearsal is lunar — philosophy follows arithmetic.
- **The test card:** whole-ship integration under real flight — the one test the ground cannot give.
- **The departure route:** Moon, home, via Mars — full-dress ops at the far planet before the long dark.
- **The rocks:** send the rocks, keep going — sample return without a course change.
- **The crater mirror:** the only sun at the shadowed crater — the Archimedes array's hardest posting.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- 225's rule rides at fleet scale: the plan is never finished — the shakedown doctrine is that rule flown as a route: test, fix, deeper test, go.

- 229's sprint number does the arithmetic: 30,000 mph (13.4 km/s) turns the Moon into a day trip and Mars into 47 days — the doctrine stands on the sling's own figure.
- The departure route closes the program's first arc: Moon shakedown, home, via Mars — and the crater mirror gives 258's Archimedes array its posting before the foundry is even built.
- The 'we are not colonizing Mars' line is seated verbatim as mission law — the ultimate reality test comes first, and the file records the order in his words.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 34 (17 August 2026 — phases 241-245, cross-checked in sitting 387), SEATED. The grade, verbatim from the batch: '244 → PASS as a test philosophy / trajectory validation.' The remaining work named by the verdict itself: trajectory validation.

§6 — BUILD SEQUENCE

- 1. Write the test card: every lander, every drill, every system — the integration list the ground cannot run.
- 2. Fly the day trip: whole ship to the Moon at sprint; the crew worked as a crew; every failure logged, none hidden.
- 3. Come home and fix: the home-and-fix leg is doctrine, not defeat — the four-beat loop's second beat.
- 4. Run the departure route: via Mars for the second full-dress rehearsal; the rocks thrown home; the ship keeps going.
- 5. Validate the trajectory: the vet's remaining item — the numbers flown against the numbers seated.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 225 (the plan is never finished):** the rule at fleet scale — the shakedown is the four-beat loop flown as a route.
- **Phase 229 (the sling):** the sprint figure the arithmetic stands on — 30,000 mph makes the Moon a day trip; the same sling throws the Moon shots at the 378 m/s mild-steel cap.
- **Phase 258 (water from rocks):** the crater mirror's posting — the Archimedes array's two-beam doctrine starts as the only sun at a shadowed crater.
- **Phase 253 (the torpedo pod):** the service-station law — the Moon's shallow well and free rocks make it the Highway's first service station; the shakedown is its first customer.

§8 — DO-NOT-BUILD

- Do NOT plan the Mars colony — the doctrine is seated: not the mission; the ultimate reality test comes first.
- Do NOT rehearse at Mars distance — 47 days one-way is not a shakedown, it is the voyage itself; the Moon or nothing.
- Do NOT partial-dress it — whole ship, all equipment, down to the landers; a rehearsal that leaves the landers home is a cruise.
- Do NOT turn the ship for the rocks — send the rocks, keep going; sample return rides the throw, never the course.
- Do NOT cap the departure at one lap — Moon, home, via Mars; the second lap is the doctrine, not an option.

§9 — OWED

OWED: the trajectory validation the vet named — the shakedown route flown against the seated numbers: 8-hour lunar transfer at sprint, the Mars-leg timing, the sling-at-Moon shot within the 378 m/s mild-steel cap, and the test card signed line by line.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-244, v1.0 — CURRENT — PASS AS A TEST PHILOSOPHY / TRAJECTORY VALIDATION (VET BATCH 34), seated law, master v2.1 (numerical print order). The two hundred and forty-fifth manual of the program — 245 of 290. Built 24 August 2026, fifth of the 20-manual 'ok next 20' shift (the author's order, 24 August 2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'. The phase's own chain, all seated in the master: the contents line and the bodies (as seated); the live instruction of 10 August 2026 (seated per sitting 147, PARTIAL — the surviving fragment) — the doctrine, verbatim: 'We are not colonizing mars, this ship is the ultimate reality test

before that is ever attempted'; the Moon day-trip arithmetic (about 8 hours at sprint; Mars 47 days); Amendment 1 (the departure route — Moon, home, via Mars; send the rocks keep going; the crater mirror; the sling's 378 m/s mild-steel cap), verbatim; the vet batch 34 grade (PASS as a test philosophy / trajectory validation) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the flown trajectory data from the day trip, or his word.

END OF MANUAL — PHASE 244